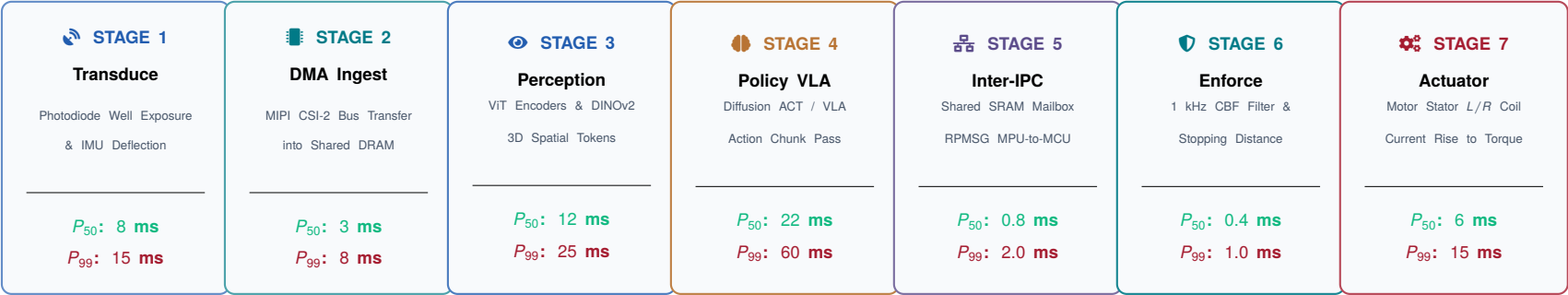




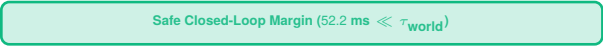
THE 7-STAGE SENSE-TO-ACTUATION LATENCY WATERFALL

Where Does the Time Go? Deconstructing the Physical, Memory, Neural, and Silicon Bottlenecks



✓ **Nominal Path** ($P_{50} = 52.2$ ms):

⚠ **Tail Path** ($P_{99} = 126.0$ ms):



World Deadline $\tau_{\text{world}} = 100$ ms